

Solve the following questions. (6×5=30)

i. What is barrier potential and how it is created in a junction diode?

Barrier potential:-

The recombination of free and mobile holes and electrons produce the narrow region at the junction called depletion region.

Depletion region contain no free and mobile charge carriers but only fixed and immobile ions. Hence this layer behaves like insulator.

Because of depletion layer, an electric potential is established across junction even when the junction is not connected to any external source of emf. It is called junction or barrier potential.

It stops further flow of carriers across the junction unless energy is supplied by an external source. Barrier voltage depends upon doping density, electronic charge and temperature.

A typical value of barrier potential for silicon is 0.7V and for germanium is 0.3V.

How it is created :-

A barrier potential is created in a junction diode by the formation of a depletion layer, which occurs when a p-type semiconductor material is brought into contact with an n-type semiconductor material.

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ii. What is zener diode? Why do we use it?

It is the diode which is designed to operate in breakdown region. The diode consists of a special, heavily doped pn-junction, designed to conduct in the reverse direction when a certain specified voltage is reached. The zener diode has a well-defined reverse breakdown voltage, at which it starts conducting current and continues operating continuously in the reverse bias without getting damaged. Additionally, the voltage drop across the diode remains constant over a wide range of voltages, a feature that makes zener diode suitable for use in voltage regulation.

Symbols:-



Uses:-

1. Zener diodes are used as voltage regulators.
2. Zener diodes are used as a fixed reference voltage in a network for biasing and comparison purposes and for calibrating voltmeters.
3. Zener diodes are used as peak clippers or voltage limiters.
4. These diodes are used for meter protection against damage from accidental application of

excessive voltage-

5. Zener diodes are used for reshaping a waveform-

iii. What are clamper circuits? Describe the working of a positive clamper-

An electronic circuit which is used to change the positive peak or negative peak of the input signal to a definite value by shifting the entire signal up or down to obtain the output signal peaks at desired levels is called clamper circuit-

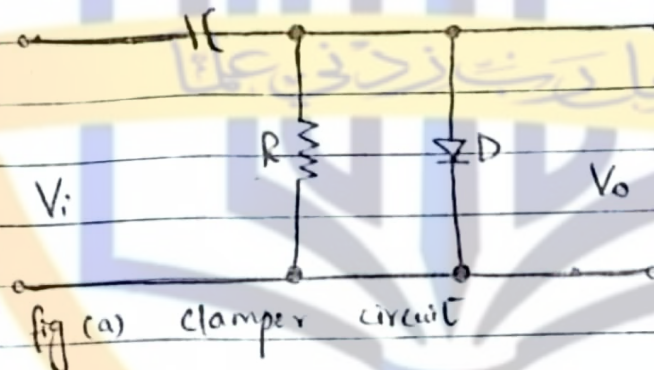
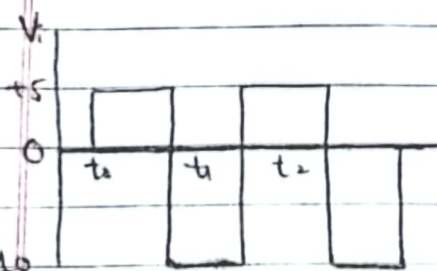
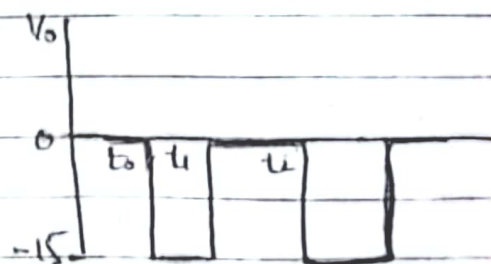


Figure (a) shows a clamper circuit consisting of diode, a resistance and a capacitor which is used for unsymmetrical input pulse at fig (a) to be clamped with its positive peaks at zero in fig (b)



Input pulse wave



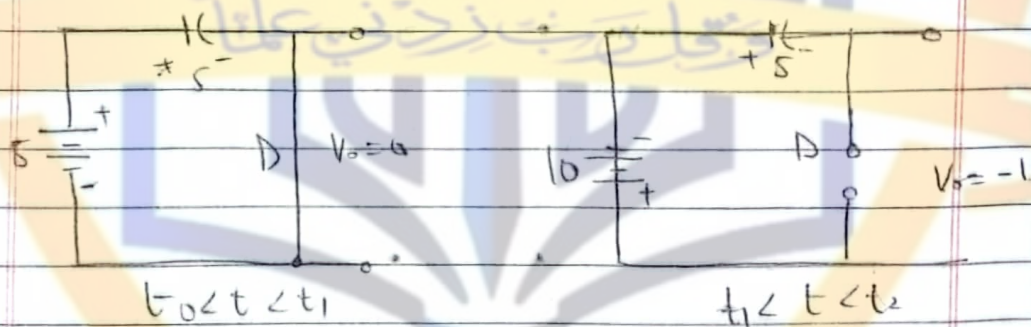
output wave clamped at 0V.

(b)

9

DATE ___/___/___

We choose the value of RC used in the circuit. So that the time constant is at least 5 times the period of input wave. When voltage goes to $+5V$ at t_0 , the capacitor charges quickly through the diode to $+5V$ and we have the circuit condition of fig (b). that exists to t_2 , the input drops to $-10V$ and with the capacitor charged to $+5V$ the diode voltage is $-15V$ and it opens, leading to the first circuit at fig (c) and $-15V$ output.



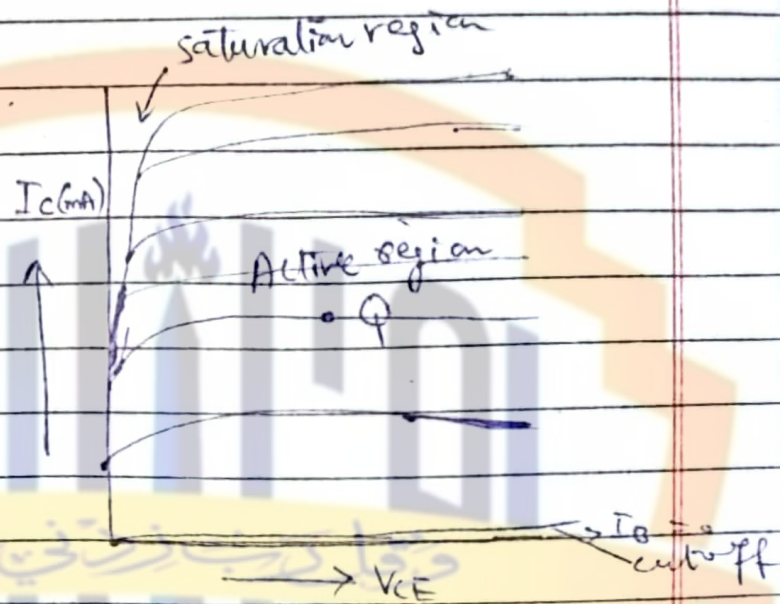
The output wave is shown in fig (b).

- iv Draw the transistor output characteristics curves. What bias conditions must exist for a transistor to operate as an amplifier.

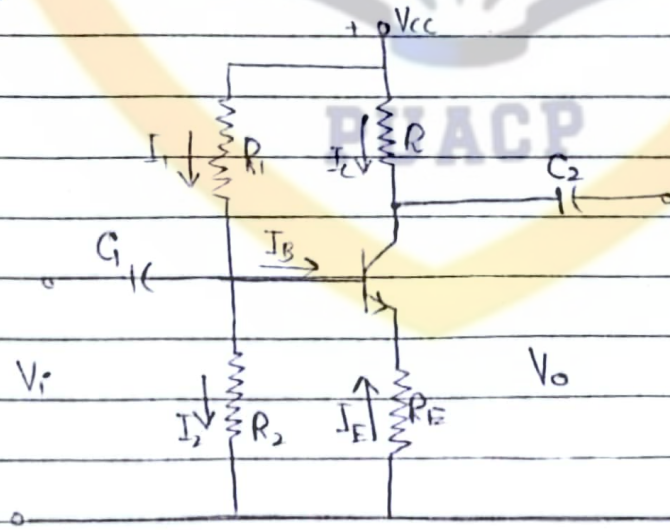
The bias conditions of the normal operation of a transistor to operate as an amplifier are that the emitter-base junction should be forward bias

DATE: ___/___/___

And the collector based junction should be reverse biased. The collector is used for the collection of charge from base, so it is reverse biased.



v. Draw the circuit diagram of a common-emitter four resistor bias amplifier and explain the role of emitter resistor R_E .



The emitter resistance provides stabilization. The voltage drop across R_E produces a voltage

6

DATE: ___/___/___

V_E with polarity as shown in fig. This voltage makes the emitter positive w.r.t base -

R_E helps stabilize a stable DC operating point for the transistor by providing negative feedback.

R_E limits the emitter current, preventing excessive current flow and potential damage to the transistor.

By providing a path for emitter current to flow, R_E helps reduce distortion and improves the amplifier's linearity.

R_E 's presence helps reduce the amplifier's sensitivity to temperature changes, β variations and other external factors.

vi. How does the fixed-bias amplifier circuit stabilize against temperature-caused changes in V_{BE} ?

This is the simplest form of the bias circuit. It provides the stabilization against the change in V_{BE} when V_{BE} changes due to change in ambient temperature.

But this circuit does not provide the stabilization when the transistor is replaced by another transistor having different value of h_{FE} . Thus this circuit is the least satisfactory in compensation for changes in h_{FE} . It can however effectively reduce bias current changes due to temperature.

DATE: ___/___/___

variation of V_{BE} -

Long Questions:-

Q#1(a)

Discuss the circuit operation of a capacitor filter when it is connected to a full-wave bridge rectifier circuit and determine its ripple factor. (6)

From Ilmi Book Page # 30.

(b).

Determine the Thevenin equivalent resistance by using the concept of active one-port network. (4)

From Ilmi Book page # 57

Q#2(a)

Draw the circuit for the common emitter (CE) transistor amplifier and find out expression for its voltage gain & current gain. (6)

From MBD page # 68

From Ilmi Book page # 81, 82, 83.

(b)

Describe BJT base voltage feedback bias amplifier circuit. (4)

From Ilmi book page # 113

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Q#3 (a)

Draw the symbol of n-channel JFET. Why it is called a unipolar transistor? (6)

From Ilmi pg # 127, 128

(b)

Describe the operation of n-channel MOSFET in enhancement mode. (4)

From Ilmi page # 131

From MBD page # 97

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